TankGuard

Technical Specification & Invention Disclosure Summary

Document Ref: TG-SPEC-2026-001 · October 2026 · IoT Water Tank Monitoring & Automated Sediment-Flush System

1. Title of Invention

A Self-Powered IoT System for In-Tank Water Quality Monitoring and Automated Sediment-Flush of Domestic and Institutional Water Storage Tanks.

Full title: "*A self-powered IoT-based in-tank water quality monitoring and automated sediment-flush system with edge-computed composite scoring, closed-loop feedback and a local-network dashboard for domestic and institutional water storage tanks.*"

2. Abstract

Water storage tanks in homes, schools, hospitals and apartment buildings degrade silently: disinfectant residual dissipates within 24–48 hours, suspended particles settle into a sludge layer at the tank floor, biofilm forms on internal surfaces, and algae proliferate in warm conditions. By the time degradation is detected by smell, taste or sight, the water is already unsafe. Existing solutions either monitor without acting, or clean by draining and scrubbing the entire tank — wasting water and labour.

TankGuard is a single, self-powered IoT unit that is mounted on the tank lid and continuously measures five water-quality parameters — **turbidity, TDS, pH, temperature and water level** — through submerged probes. An on-device ESP32 microcontroller performs all computation at the network edge: it fuses the readings into a weighted composite **TankScore**, analyses 14-day trends, and classifies the contamination event (gradual biofilm growth, source-water contamination, or sludge resuspension). When the score indicates that settled contamination has breached a threshold, the unit actuates a normally-closed valve at the tank's lowest drain point, selectively discharging the sediment-rich bottom layer *without draining the bulk water column*. After flushing, the system re-measures water quality and computes a **flush-effectiveness metric**, closing a verification loop absent from prior art. All logic runs locally with no cloud, subscription or internet requirement; a Wi-Fi access point and embedded web server present an interactive dashboard to any phone or browser on the local network. The unit is powered by a small solar panel and battery, with duty-cycled sensing for multi-week autonomy.

3. Components Used

TankGuard is built entirely from commercially available off-the-shelf (COTS) components. The table below lists the components of a working embodiment, with a representative part and its function.

Component	Representative Part	Interface	Function
Microcontroller (edge processor)	ESP32-WROOM-32 DevKit	Wi-Fi, GPIO, I²C, 1-Wire, ADC	Runs all firmware — sampling, TankScore computation, trend regression, event classification, flush decision, embedded HTTP server, mDNS, flash logging and deep-sleep power management
ADC	ADS1115, 16-bit, 4-channel	I²C @ 0x48	High-resolution digitisation of the pH electrode and analog sensors; ESP32's internal ADC is too coarse for millivolt-level pH
Turbidity sensor	DFRobot SEN0189 (or generic clone)	Analog	90° infrared nephelometry — primary indicator of suspended solids, biofilm fragments and sludge resuspension (40% of TankScore)
TDS sensor	Gravity SEN0244 / XKC-Y25-NPN	Analog + NTC	Two-electrode conductivity with temperature compensation — dissolved-solids leaching from settled debris (25% of TankScore)
pH sensor	Gravity SEN0161-V2 / glass electrode + front end	Analog via ADS1115	Potentiometric pH — indicates algal activity, anaerobic decay and corrosion (20% of TankScore)
Temperature probe	DS18B20 waterproof (stainless)	1-Wire	Temperature compensation plus independent biological-activity indicator (5% of TankScore)
Water-level sensor	JSN-SR04T waterproof ultrasonic	Trigger / Echo GPIO	Water volume and refill detection; enables discrimination of source contamination from sludge resuspension (10% of TankScore)
Flush valve	1″ brass normally-closed solenoid, 12 V DC	2-wire, MOSFET-switched	The actuator — opens on command to discharge the sediment-rich bottom layer through the tank's lowest drain point
Valve driver	N-channel MOSFET (IRLZ44N) + flyback diode	Gate ← ESP32 GPIO	Isolates 3.3 V logic from the 12 V coil; provides inductive-kickback protection and timed control
Power system	10 W solar panel, 4×18650 Li-ion, CN3791 MPPT, LM2596 buck + AMS1117	—	Self-sustaining off-grid power with 2–3 weeks of zero-sun autonomy; regulated 12 V / 5 V / 3.3 V rails
Enclosure	IP66 ABS/polycarbonate box + IP68 cable glands + heat shield	—	Weatherproof, heat-protected housing mounted on the tank lid
Calibration set	pH 4.0 / 7.0 buffer sachets + distilled water + KCl storage solution	—	Two-point pH calibration and probe conditioning
Firmware	ESP-IDF / Arduino framework + LittleFS	Wi-Fi AP/STA	All edge logic and the local dashboard (software-implemented portion)

4. Features

- **Multi-parameter sensing** — turbidity, TDS, pH, temperature and water level measured continuously in-tank
- **TankScore composite index** — one 0–100 number answering "is it time to clean?"
- **Trend-predictive analysis** — 14-day linear regression detects degradation before visible symptoms
- **Event classification** — distinguishes biofilm growth, source-water contamination and sludge resuspension
- **Automated selective flush** — bottom-layer drain without emptying the tank
- **Closed-loop verification** — re-measures and scores after each flush, computing effectiveness
- **Manual override** — "Flush Now" and "Tank Cleaned" buttons on the dashboard
- **Solar-powered autonomy** — runs off-grid for weeks without sun
- **Cloud-free local dashboard** — any phone/browser, no internet, no account
- **Edge data logging** — full history stored on-device; optional OTA updates and alert exports

5. What Makes the System Unique

The invention is distinguished from prior art (sensor-only monitors, full-tank drain cleaners, industrial camera-based sludge systems, in-pipe point-of-use devices) by the following non-obvious combinations:

Unique Aspect	Distinguishing Feature
1. Closed-loop flush verification	The system re-measures water quality *after* flushing and computes a quantitative effectiveness score, then re-flushes or escalates if ineffective. No surveyed prior art verifies cleaning success — existing systems operate open-loop.
2. Selective bottom-layer flush (not full drain)	The valve at the tank's lowest point discharges only the sediment-enriched bottom 5–15% of volume, keeping the bulk water. Fundamentally different from full drain-and-scrub systems.
3. Trend-predictive TankScore	Triggering uses a weighted composite with 14-day slope and event classification, not a single-sensor threshold — enabling action *before* visible degradation.
4. Sludge-resuspension detection	Correlating turbidity spikes against level changes distinguishes resuspension from source contamination — a specific diagnosis absent from prior art.
5. Edge-only IoT architecture	All sensing, scoring, decision and verification execute on-device; the dashboard is self-hosted over the local network. No cloud, subscription or account — a deliberate architectural contrast to all surveyed IoT monitors.
6. Duty-cycled solar autonomy	Deep-sleep MCU with switched sensor power rails achieves multi-week off-grid operation while maintaining continuous logical monitoring.
7. Single integrated unit	Sensing, edge computing, actuation, power and user interface combined in one tank-lid-mounted device requiring only a single drain-tap connection.

6. Expected Cost

Two build tiers are available. All figures are parts-only and exclude GST, freight and installation.

Tier	Description	Parts Cost	Installed (approx.)
Full / proposal build	Grade-A Gravity sensors, IP66 enclosure, 1″ food-grade valve, full integration	**₹19,470**	~₹20,500
Minimum prototype	Generic/clone sensors, salvaged 18650 cells, basic enclosure	**~₹6,500**	~₹7,200

6.1 Full Build Breakdown

Module M1 — Core Monitor		**₹15,500**
ESP32, ADS1115, sensors, solar/battery/power, enclosure, calibration, engineering	Five-parameter monitor with solar power, edge dashboard and TankScore logic	—
Module M2 — Sludge Flush		**₹3,970**
1″ brass NC solenoid valve, PVC drain kit, SS bracket, driver board, cable, integration	Automated bottom-flush hardware with manual bypass	—
Total Parts Cost		**₹19,470**

Notes: The 1″ brass solenoid valve (₹2,500 in the full tier) is specified for high flow and reliable sludge removal — smaller ½″ valves tend to clog. The minimum tier (~₹6,500) trades sensor grade and enclosure quality for cost; readings are adequate for trend detection (±5–10%) but not for compliance-grade measurement. An optional pump-control module (M3) can be added later (~₹1,080).

7. Technical Specifications

7.1 Hardware Specifications

Parameter	Specification
Microcontroller	ESP32-WROOM-32 — dual-core Xtensa LX6 @ 240 MHz, 520 KB SRAM, 4 MB flash; Wi-Fi 802.11 b/g/n, Bluetooth LE
ADC	ADS1115 — 16-bit, 4-channel, I²C; programmable gain; internal reference
Turbidity	0–3000 NTU range (practical 0–100 NTU), IR 90° nephelometry, IP68 probe
TDS	0–1000 ppm (extendable), 3–5.5 V excitation, ±10% FS, temperature-compensated
pH	0–14 pH, glass electrode, two-point calibrated (4.0 / 7.0), ±0.1 pH after calibration
Temperature	−10 to +85 °C, ±0.5 °C (DS18B20), waterproof stainless probe
Water level	25–450 cm (JSN-SR04T), ±1 cm, IP67 ultrasonic
Flush valve	1″ normally-closed solenoid, 12 V DC, ~₹2,500 tier; brass body, EPDM seal; manual bypass valve
Valve driver	N-channel MOSFET low-side switch, flyback diode, PWM-capable; 12 V coil switched from 3.3 V logic
Power — solar	10 W, 12 V polycrystalline panel (5 W 6 V in minimum tier)
Power — battery	4×18650 Li-ion (~10 Ah @ 3.7 V equivalent); 2–3 weeks zero-sun autonomy
Charging	CN3791 MPPT charge controller with over-charge / over-discharge protection
Regulated rails	12 V (valve) · 5 V (sensor excitation) · 3.3 V (MCU/logic) via LM2596 + AMS1117
Sleep current	< 1 mA in deep sleep between measurement cycles
Enclosure	IP66 polycarbonate/ABS, heat shield, IP68 cable glands; lid-mounted
Operating temperature	0–55 °C ambient (electronics); probes −10 to +85 °C

7.2 Software Specifications

Parameter	Specification
Firmware framework	ESP-IDF (or Arduino-ESP32); modular C/C++
Measurement cycle	Wake every 3 hours (configurable); median-filtered multi-sample acquisition
TankScore	0–100 weighted composite: turbidity 40%, TDS 25%, pH 20%, level 10%, temperature 5%
Trend analysis	Linear regression over 14-day window (112 samples); slope/acceleration penalties
Event classification	3 classes — gradual degradation, source-water contamination, sludge resuspension
State machine	GOOD (80–100) → WATCH (50–79) → FLUSH RECOMMENDED (30–49) → CLEAN REQUIRED (0–29)
Flush control	Duration 30–120 s computed from tank volume, pre-flush turbidity, sludge parameter; ≤15% volume discharged
Verification	Settle 15–60 min, re-measure, compute effectiveness = (pre−post)/pre ×100; threshold 40%
Local server	Embedded HTTP server + mDNS (`tankguard.local`), AP or STA mode
Dashboard	HTML5/JS single-page — gauges, trend charts (24 h / 7 d / 30 d / 90 d), TankScore banner, flush controls, history
Storage	LittleFS on flash — ring buffer of samples + event log (pre/post scores, effectiveness)
Connectivity	Wi-Fi only by default (no cloud); optional MQTT/e-mail/SMS alerts are notification-only, never control
OTA	Optional over-the-air firmware update while installed
Power management	Deep-sleep scheduler with switched sensor power rail; <1 mA idle

8. Drawings

The following figures are schematic and not to scale. They illustrate the architecture, operating cycle, control states and physical mounting of the invention.

Fig. 1 — System Architecture (five-layer IoT stack)

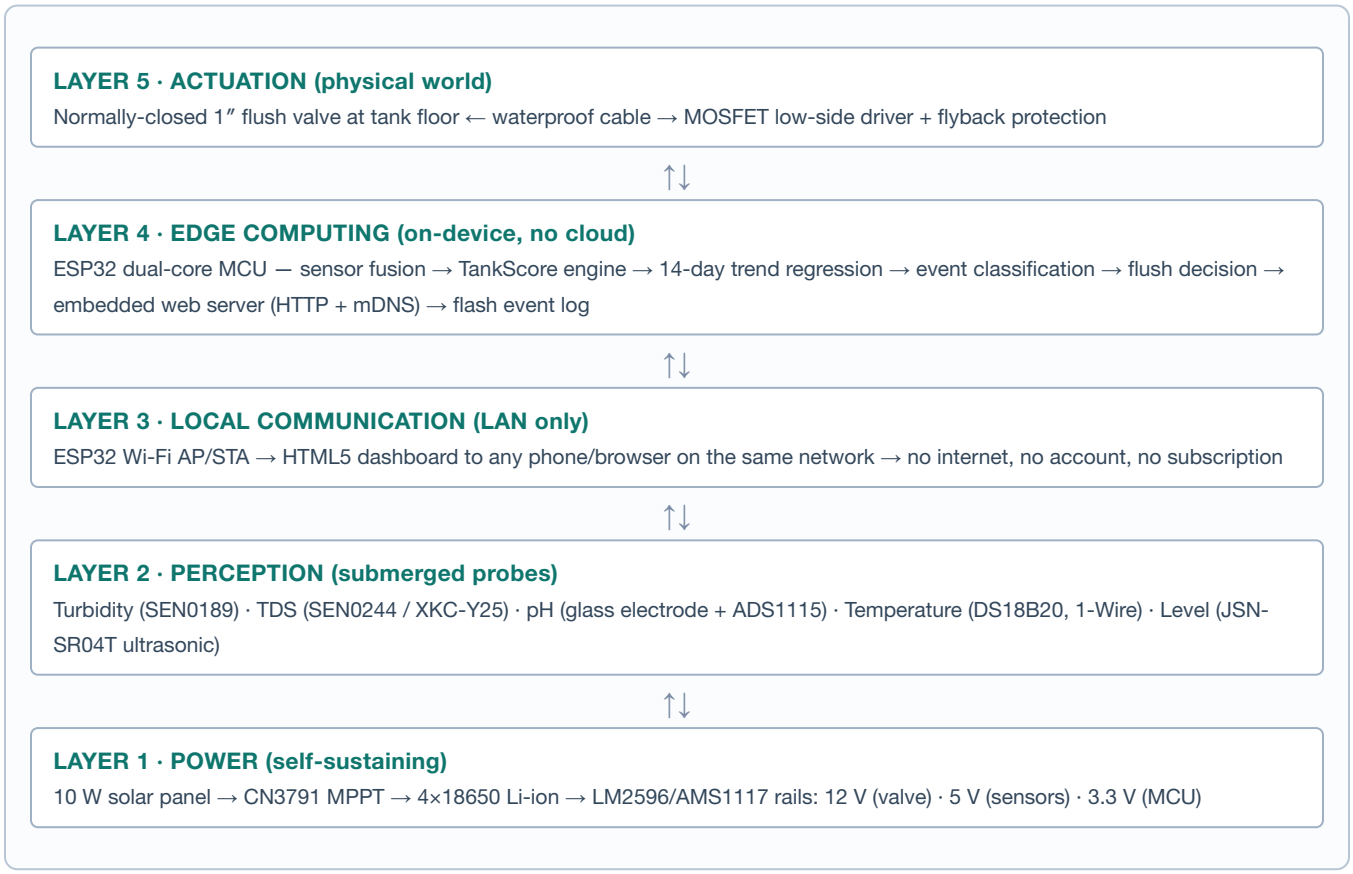


Fig. 1 — Four functional layers plus power; all computation stays at Layer 4 (edge).

Fig. 2 — Operating Cycle (sense → compute → actuate → verify)

Step 1 — Wake: RTC timer wakes ESP32 (every 3 h); sensor power rail energised.

Step 2 — Acquire: Read turbidity, TDS, pH, temperature, level; median-filter each.

Step 3 — Score: Normalise + weight parameters into TankScore; append sample to 14-day buffer.

Step 4 — Analyse: 14-day regression + classify event (degradation / source contamination / sludge resuspension).

Step 5 — Decide: Map score to GOOD / WATCH / FLUSH RECOMMENDED / CLEAN REQUIRED; issue flush command (auto or manual).

Step 6 — Flush: MOSFET opens bottom valve 30–120 s; sediment-rich bottom 5–15% discharged, bulk water retained.

Step 7 — Settle: Valve closes; 15–60 min settling interval.

Step 8 — Verify (closed loop): Re-measure; compute flush effectiveness = (pre − post)/pre × 100; re-flush or escalate if <40%.

Step 9 — Report: Log event; update local dashboard (readings, TankScore, trends, history, battery).

Step 10 — Sleep: De-energise sensors; deep sleep <1 mA until next cycle.

Fig. 2 — The closed-loop verification (Step 8) is the core novel step.

Fig. 3 — TankScore State Diagram



GOOD → WATCH → FLUSH RECOMMENDED → CLEAN REQUIRED | after verified flush, score recovers toward GOOD; if verification fails, escalate to manual cleaning.

Fig. 3 — TankScore state transitions and corresponding system actions.

Fig. 4 — Mounting Schematic (elevation, not to scale)

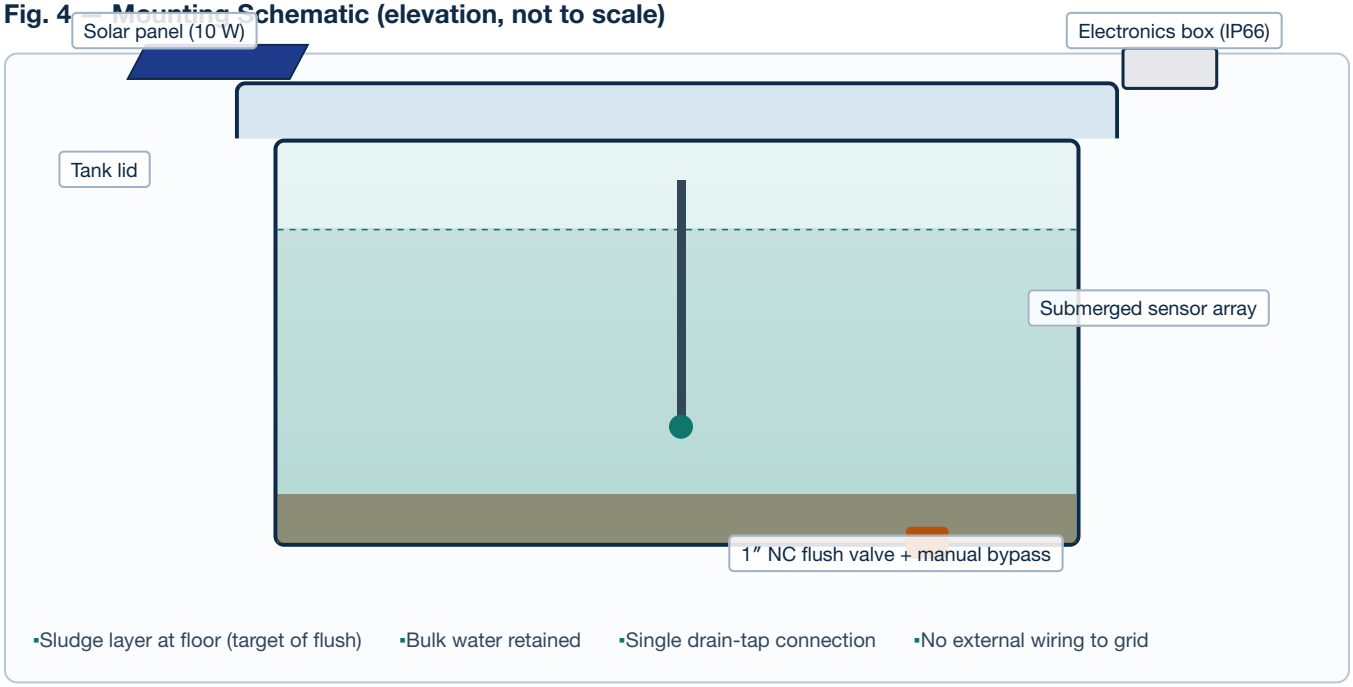


Fig. 4 — Lid-mounted unit with suspended probes and a bottom drain valve; only one plumbing connection is required.

Summary: TankGuard converts water-tank maintenance from calendar-based guesswork into evidence-based, automated care — sensing at the edge, deciding locally, flushing selectively, and verifying its own work, with no cloud dependency.